

HOMEWORK 5 SOLUTIONS

Problem 1 Use the provided dataset “HW5_data.csv” for this question. Consider the linear model

$$y = X\beta + \sigma\epsilon$$

where y is a $n \times 1$ vector of observations, X is the $n \times p$ design matrix, β is the $p \times 1$ vector of coefficients, σ^2 is the variance and ϵ is a $n \times 1$ vector of independent standard normal random variables. For the data set, $n = 200$ and $p = 5$.

- (1) (8 pts) Using the notation in the slides, matrices Q , P and Λ are such that

$$QX'XQ' = \Lambda^2 \quad PXQ' = D = \begin{pmatrix} \Lambda \\ 0 \end{pmatrix}, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$$

where the (λ_j) are the square roots of the eigenvalues of $X'X$. Show that if $\nu = P\epsilon$ then $E(\nu_i) = 0$ and $\text{Var}(\nu_i) = 1$ for all $i = 1, \dots, n$.

Solution:

P comes from the decomposition of X , so P is a constant matrix and $\epsilon \sim N(0, I)$. Hence,

$$E(\nu) = E(P\epsilon) = PE(\epsilon) = 0, \quad \text{Var}(P\epsilon) = P\text{Var}(\epsilon)P' = PIP' = I$$

- (2) (10 pts) If $\gamma = Q\beta$, show that $PX\beta$ can be written as $D\gamma$. Hence, show that the original linear model can be written as

$$z_i = \begin{cases} \lambda_i \gamma_i + \sigma \nu_i & i = 1, \dots, p \\ \sigma \nu_i & i = p+1, \dots, n, \end{cases}$$

where $z = Py$.

Solution:

$$D\gamma = PXQ'\gamma = PXQ'Q\beta = PX\beta$$

$$z = Py = P(X\beta + \sigma\epsilon) = D\gamma + \sigma\nu$$

The first p rows of D is $\text{diag}(\lambda_1, \dots, \lambda_p)$, and the other $n - p$ rows of D are all 0, so z_i can be written in this way.

- (3) (10 pts) Prove that

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{j=1}^p \frac{1}{\lambda_j^2}.$$

(Hint, the trace of a matrix is the sum of its eigenvalues.)

Solution:

$\text{Var}(\hat{\beta}_j) = \sigma^2 C_j$, where C_j is the j th diagonal element of $(X'X)^{-1}$.

Therefore, the eigenvalues of $X'X$ are (λ_j^2) , so the eigenvalues of $(X'X)^{-1}$ are $(1/\lambda_j^2)$.

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_j) = \sigma^2 \sum_{j=1}^p C_j = \sigma^2 \text{Trace}[(X'X)^{-1}] = \sigma^2 \sum_{j=1}^p \frac{1}{\lambda_j^2}$$

(4) (10 pts) From the data set provided, find $\hat{\gamma}$.

Solution:

First, do the SVD decomposition (can use the `svd` function in R):

$$X = P'DQ$$

Then, transform:

$$z = Py = PX\beta + \sigma P\epsilon = D\gamma + \sigma\nu$$

Then $\hat{\gamma}_j = z_j/\lambda_j$ for $j = 1, \dots, 5$, where (λ_j) are the square root of the eigenvalues of $X'X$.

$$\hat{\gamma} = (0.884, 0.064, 1.662, -1.414, 0.305)$$

$$\text{or } \hat{\gamma} = (0.884, 0.064, -1.662, 1.414, 0.305)$$

(5) (12 pts) Now suppose we want to keep m components where m is the smallest integer for which

$$\frac{\sum_{j=1}^m \lambda_j^2}{\sum_{j=1}^p \lambda_j^2} > 0.9.$$

Find m and the corresponding $\hat{\beta}_{pca}$

Solution:

Sort (λ_j) by descending order,

$$\frac{\sum_{j=1}^3 \lambda_j^2}{\sum_{j=1}^p \lambda_j^2} < 0.9, \quad \frac{\sum_{j=1}^4 \lambda_j^2}{\sum_{j=1}^p \lambda_j^2} > 0.9, \quad \text{so } m = 4.$$

Drop the 5th row of Q and the 5th element of $\hat{\gamma}$,

$$\hat{\beta}_{pca} = Q'_{red} \hat{\gamma}_{red} = (1.570, -0.967, 0.117, 0.057, -1.459)$$

Problem 2

Consider the Bayesian linear model

$$y_i \sim N(x_i\beta, \sigma^2), \quad i = 1, \dots, n,$$

$$\text{where } \sum_{i=1}^n x_i = 0, \quad \sum_{i=1}^n x_i^2 = n \quad \text{and} \quad \sum_{i=1}^n x_i y_i = \gamma.$$

The prior for β is given by

$$\pi(\beta) = 0.5 \mathbf{1}(\beta = 0) + 0.5 N(\beta|0, \tau^2),$$

where $\mathbf{1}(\beta = 0)$ is the probability mass function which puts all probability on $\beta = 0$.

Suppose σ and τ are known.

- (1) (8 pts) Write down the posterior density for β and note that this is always of the form prior times likelihood.

Solution:

$$\pi(\beta | y) \propto p(y | \beta) \pi(\beta) \propto \prod_{i=1}^n N(y_i | x_i \beta, \sigma^2) [\mathbf{1}(\beta = 0) + N(\beta | 0, \tau^2)]$$

or

$$\pi(\beta | y) \propto p(y | \beta) \pi(\beta) \propto \exp \left\{ -\frac{1}{2} \sum_{i=1}^n (y_i - x_i \beta)^2 / \sigma^2 \right\} \left\{ \mathbf{1}(\beta = 0) + \frac{1}{\tau \sqrt{2\pi}} \exp \left(-\frac{1}{2} \beta^2 / \tau^2 \right) \right\}$$

- (2) (12 pts). By integrating out β , find the marginal likelihood

$$m(y_1, \dots, y_n) = \int \prod_{i=1}^n N(y_i | x_i \beta, \sigma^2) N(\beta | 0, \tau^2) d\beta.$$

Solution:

Consider $p(y, \beta) = p(y | \beta) p(\beta) = \prod_{i=1}^n N(y_i | x_i \beta, \sigma^2) N(\beta | 0, \tau^2)$ as a joint distribution of β and $y = (y_1, \dots, y_n)$. To integrate out β and find the marginal likelihood $m(y)$, we can rearrange the equation as $p(y, \beta) = m(y) p(\beta | y)$, where $p(\beta | y)$ should be a normal distribution.

$$\begin{aligned} & \int \prod_{i=1}^n N(y_i | x_i \beta, \sigma^2) N(\beta | 0, \tau^2) d\beta \\ &= \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^n \frac{1}{\tau \sqrt{2\pi}} \int \exp \left\{ -\frac{1}{2} \sum_{i=1}^n (y_i - x_i \beta)^2 / \sigma^2 - \frac{1}{2} \beta^2 / \tau^2 \right\} d\beta \end{aligned}$$

(First, move the terms without β out of the integral.)

$$= \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^n \frac{1}{\tau \sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n y_i^2 / \sigma^2 \right\} \int \exp \left\{ -\frac{1}{2} \left[\left(\frac{1}{\tau^2} + \frac{1}{\sigma^2} \sum_{i=1}^n x_i^2 \right) \beta^2 - \frac{2}{\sigma^2} \sum_{i=1}^n x_i y_i \beta \right] \right\} d\beta$$

(Then, complete the square for the β terms to make it into the form of a normal density.)

$$= \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^n \frac{1}{\tau \sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \sum_{i=1}^n \frac{y_i^2}{\sigma^2} + \frac{1}{2} \frac{\gamma^2 / \sigma^4}{n / \sigma^2 + 1 / \tau^2} \right\} \int \exp \left\{ -\frac{1}{2} \left(\frac{1}{\tau^2} + \frac{n}{\sigma^2} \right) \left(\beta - \frac{\gamma / \sigma^2}{n / \sigma^2 + 1 / \tau^2} \right)^2 \right\} d\beta$$

(Finally, find the normalizing constant for the normal density and solve the integral by using the fact that the integral of a density is always 1.)

$$= \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^n \exp \left\{ -\frac{1}{2} \sum_{i=1}^n y_i^2 / \sigma^2 \right\} \exp \left\{ \frac{1}{2} \frac{\gamma^2 / \sigma^4}{n / \sigma^2 + 1 / \tau^2} \right\} \frac{1 / \tau}{\sqrt{n / \sigma^2 + 1 / \tau^2}},$$

- (3) (12 pts) Hence, show that

$$\pi(\beta | y_1, \dots, y_n) \propto \left\{ \prod_{i=1}^n N(y_i | 0, \sigma^2) \right\} \mathbf{1}(\beta = 0) + m(y_1, \dots, y_n) N(\beta | \nu, \psi^2)$$

and find ν and ψ .

Solution:

In Problem 2 (2), we rearrange

$$\prod_{i=1}^n N(y_i | x_i \beta, \sigma^2) N(\beta | 0, \tau^2)$$

as $m(y)p(\beta | y)$, where

$$p(\beta | y) \propto \exp \left\{ -\frac{1}{2} \left(\frac{1}{\tau^2} + \frac{n}{\sigma^2} \right) \left(\beta - \frac{\gamma/\sigma^2}{n/\sigma^2 + 1/\tau^2} \right)^2 \right\}$$

is a normal distribution, so

$$\prod_{i=1}^n N(y_i | x_i \beta, \sigma^2) N(\beta | 0, \tau^2) = m(y_1, \dots, y_n) N(\beta | \nu, \psi^2)$$

with

$$\nu = \frac{\gamma/\sigma^2}{n/\sigma^2 + 1/\tau^2}, \quad \psi^2 = \left(\frac{1}{\tau^2} + \frac{n}{\sigma^2} \right)^{-1}$$

(4) (8 pts) Hence, write down $\pi(\beta = 0 | y_1, \dots, y_n)$.

Solution:

$$\pi(\beta = 0 | y_1, \dots, y_n) \propto \prod_{i=1}^n N(y_i | 0, \sigma^2) + m(y_1, \dots, y_n) N(\beta = 0 | \nu, \psi^2)$$

or

$$\pi(\beta = 0 | y_1, \dots, y_n) \propto \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^n \exp \left\{ -\frac{1}{2} \sum_{i=1}^n y_i^2 / \sigma^2 \right\} + m(y_1, \dots, y_n) \frac{1}{\psi\sqrt{2\pi}} \exp \left(-\frac{\nu^2}{2\psi^2} \right)$$

or

$$\pi(\beta = 0 | y_1, \dots, y_n) \propto \exp \left(-\frac{1}{2} \sum_{i=1}^n y_i^2 / \sigma^2 \right) \left(1 + \frac{1}{\tau\sqrt{2\pi}} \right)$$

(5) (10 pts) What is the significance of the term

$$\frac{m(y_1, \dots, y_n)}{\prod_{i=1}^n N(y_i | 0, \sigma^2)}$$

and show that this converges to infinity as γ^2 grows.

Solution:

$$\frac{m(y_1, \dots, y_n)}{\prod_{i=1}^n N(y_i | 0, \sigma^2)} = \exp \left\{ \frac{1}{2} \frac{\gamma^2 / \sigma^4}{n / \sigma^2 + 1 / \tau^2} \right\} \frac{1 / \tau}{\sqrt{n / \sigma^2 + 1 / \tau^2}}.$$

This term is the ratio between the marginal likelihood of y when assuming $\beta \sim N(0, \tau^2)$ and that when assuming $\beta = 0$.

Or, this term is the Bayes factor comparing the model with $\beta \sim N(0, \tau^2)$ and the model with $\beta = 0$.

When $\gamma^2 \rightarrow \infty$, the form of the ratio is easily seen to go to infinity, supporting the model with $\beta \sim N(0, \tau^2)$. This makes sense since $\gamma^2 \rightarrow \infty$ is not compatible with the y_i being normal with zero mean and finite variance.